

Notes: Baghai, Silva, and Ye (RFS, 2024)

Teams and Bankruptcy

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1 Big picture

1.1 Main question

Bankruptcy reallocates resources, but it can also destroy organizational capital. This paper focuses on one specific component: **human capital embedded in teams**. The question is:

When a firm goes bankrupt, do inventor teams break apart, and does the resulting loss of team-specific human capital reduce innovation productivity?

1.2 Core mechanism

The mechanism is a simple complementarity argument:

- An inventor’s productivity depends not only on individual skill, but also on *who they work with*.
- Team-specific human capital forms over repeated collaboration (shared routines, communication, division of labor).
- Bankruptcy increases frictions to keeping teams intact (financial constraints, coordination costs, matching frictions, local shocks).
- If teams dissolve, productivity may fall *especially* for inventors whose pre-bankruptcy production relied heavily on teamwork.

1.3 Empirical strategy in one sentence

Estimate the change in inventor productivity before vs. after bankruptcy (staggered events), then test whether the productivity drop is **larger for team-dependent inventors**, and link that differential drop to evidence on team disruption and team reallocation.

2 Data and measurement

2.1 Unit of observation and timeline

The primary panel is at the inventor–firm–year level (i, f, t) :

- inventors are linked to employers (patent assignees/firms),
- productivity is measured using patents filed in a given year and subsequent citations,
- bankruptcy events occur at different times across firms (1980–2015 in the sample).

2.2 Key outcome variables (inventor productivity)

The paper uses multiple measures; the two main ones are:

$$\ln(\text{Patents}_{ift} + 1), \quad \ln\left(\frac{\text{Citations}_{ift}}{\text{Patents}_{ift}} + 1\right), \quad (1)$$

where citations are forward citations to patents filed in year t .

Additional outcomes used in robustness:

$$\ln(\text{Citations}_{ift} + 1), \quad \ln(\text{DollarValue}_{ift} + 1). \quad (2)$$

2.3 Treatment timing

Define a bankruptcy filing year for each firm f . Let:

$$Post_{it} \equiv \mathbf{1}\{\text{year } t \text{ is after inventor } i\text{'s associated firm's bankruptcy}\}.$$

The analysis often replaces $Post_{it}$ with event-time dummies $t0, t1, t2, \dots$ relative to the filing year.

2.4 Team dependence and team-specific human capital

The key heterogeneity measure is **Team dependence**, an inventor-level constant:

For inventors employed by a bankrupt firm in at least one of the three years before the filing, **Team dependence** is the share of that inventor's pre-bankruptcy patents that are coauthored with other inventors also employed at that firm.

It ranges from 0 (no within-firm coauthorship) to 1 (all output coauthored with within-firm teammates).

2.5 Team stability variable

To measure whether teams remain intact, the paper defines:

- Construct inventor i 's *team* as the set of inventors who are *both* coworkers and coauthors of i in year $t - 4$.
- Compute the share of these teammates still employed at the same firm in year t .
- Define **Remain together** $_{ift}$ as an indicator equal to 1 if this share exceeds 75%.

2.6 Pair-level team measures (for joint mobility)

For inventor pair (i, j) within the same pre-bankruptcy firm:

- **Pair codependence**: share of the pair's patents coauthored by both inventors (up to year $t - 1$).
- **Pair average citations**: mean citations of the pair's jointly authored patents (up to year $t - 1$).

3 Models: all regression specifications used in the paper

3.1 Model 1: Baseline bankruptcy effect on inventor productivity (Equation (1))

The baseline DID regression is:

$$Productivity_{ift} = \alpha + \beta Post_{it} + \Gamma' \mathbf{Z}_{ift} + \varepsilon_{ift}. \quad (3)$$

Here $Productivity_{ift}$ is alternately $\ln(\text{Patents} + 1)$ or $\ln(\text{Citations per patent} + 1)$. The control matrix $\mathbf{Z}_{ift}$ includes:

- inventor fixed effects (absorbing time-invariant inventor and firm differences),
- experience fixed effects (dummies for years since first observed patent),
- technology-class-by-year fixed effects (absorbing shocks to fields over time).

Interpretation of β . β is the average within-inventor change in productivity after bankruptcy, net of technology-time effects and inventor life-cycle patterns.

3.2 Model 1b: Dynamic event-study version (Table 3, Panel A)

Replace $Post_{it}$ with event-time indicators:

$$Productivity_{ift} = \alpha + \sum_{k \in \mathcal{K}} \delta_k \mathbf{1}\{t = t_k\} + \Gamma' \mathbf{Z}_{ift} + \varepsilon_{ift}, \quad (4)$$

where $t_0, t_1, t_2, t_3, t_4, t_5$ plus correspond to specific years relative to the filing.

3.3 Model 1c: Skill portability and adjustment channels (Table 3, Panel B)

To test whether inventors with more portable skills are less harmed, the paper interacts post-bankruptcy with:

- **Change tech. class:** switching main technology class after bankruptcy;
- **Past generality:** pre-bankruptcy patent generality (broader applicability).

A representative specification is:

$$Productivity_{ift} = \alpha + \beta Post_{it} + \phi (Post_{it} \times Portability_i) + \Gamma' \mathbf{Z}_{ift} + \varepsilon_{ift}. \quad (5)$$

A positive ϕ means portable skills mitigate post-bankruptcy productivity losses.

3.4 Model 2: Team dependence heterogeneity (Equation (2))

The key team-specific-human-capital test is:

$$Productivity_{ift} = \alpha + \beta_1 Post_{it} + \beta_2 (Post_{it} \times TeamDep_i) + \Gamma' \mathbf{Z}_{ift} + \varepsilon_{ift}. \quad (6)$$

Compared to (3), the control set includes bankrupt-firm fixed effects (and in some variants bankrupt-firm-by-year fixed effects), plus the same experience and technology-class-by-year fixed effects, and errors clustered at the bankrupt firm level.

Key marginal effect. For an inventor with $TeamDep_i = d \in [0, 1]$, the post-bankruptcy effect is:

$$\left. \frac{\partial Productivity}{\partial Post} \right|_{TeamDep=d} = \beta_1 + \beta_2 d.$$

Thus, $\beta_2 < 0$ is direct evidence that productivity drops more for inventors who relied more on teamwork.

3.5 Model 2b: Dynamic team-dependence event study (Figure 5; Table 5, Panel B)

A dynamic interaction version replaces $Post_{it}$ with event-time dummies interacted with $TeamDep_i$:

$$Y_{ift} = \alpha + \mathbf{T}'_{it} \beta + (TeamDep_i \cdot \mathbf{T}'_{it}) \theta + \Gamma' \mathbf{Z}_{ift} + \varepsilon_{ift}, \quad (7)$$

where $\mathbf{T}_{it}$ includes dummies for many years before and after bankruptcy; the interaction for $t - 1$ is omitted as the baseline.

3.6 Model 3: Team stability around bankruptcy (Table 4)

Let $RemainTogether_{ift}$ be the team stability indicator. The baseline event-time regression is:

$$RemainTogether_{ift} = \alpha + \sum_{k \in \{0, 1, 2, 3+\}} \pi_k t_k + \Gamma' \mathbf{Z}_{ift} + u_{ift}, \quad (8)$$

with t_0, t_1, t_2, t_{3plus} defined in the paper.

To study how team stability depends on **bankruptcy outcomes** (liquidation, independent exit, acquisition by strategic/financial buyer) or **mobility restrictions** (noncompete enforceability), the paper estimates interacted versions:

$$RemainTogether_{ift} = \alpha + \sum_k \pi_k t_k + \sum_k \pi_k^{(A)} (t_k \times A_f) + \Gamma' \mathbf{Z}_{ift} + u_{ift}, \quad (9)$$

where A_f is an indicator for a particular post-bankruptcy outcome or state-level regime.

3.7 Model 4: Robustness variations (Table 6)

Table 6 implements several robustness changes to (6), including:

- alternative samples (star inventors; low-R&D firms; all inventors including never-treated),
- alternative dependent variables (citations; dollar value; levels via Poisson),
- additional controls (tenure at bankrupt firm and its interaction with $Post_{it}$),
- alternative functional forms (Poisson count model for patents).

3.8 Model 5: Career continuation / selection (Table 7)

To study whether team dependence predicts exiting inventive activity after bankruptcy, the paper uses an inventor-level linear probability model:

$$StillInventing_i(x) = \alpha + \lambda TeamDep_i + \Gamma' \mathbf{W}_i + \varepsilon_i, \quad (10)$$

where $StillInventing_i(x)$ is 1 if inventor i files any patent within x years after the bankruptcy.

3.9 Model 6: Heterogeneity by bankruptcy resolution (Table 8)

To test which resolutions preserve team capital, the paper augments (6) with interactions between $(Post \times TeamDep)$ and bankruptcy outcomes:

$$Productivity_{ift} = \alpha + (Post_{it} \times TeamDep_i) \beta + \sum_m (Post_{it} \times TeamDep_i \times Outcome_f^m) \beta_m + \Gamma' \mathbf{Z}_{ift} + \varepsilon_{ift}. \quad (11)$$

3.10 Model 7: Mobility restrictions (Table 9)

Let $StrongNCA_s$ indicate strong noncompete enforceability in the inventor's state at the time of bankruptcy. A triple-interaction version is:

$$\begin{aligned} Productivity_{ift} = & \alpha + \beta(Post_{it} \times TeamDep_i) + \psi(Post_{it} \times StrongNCA_s) \\ & + \omega(Post_{it} \times TeamDep_i \times StrongNCA_s) + \Gamma' \mathbf{Z}_{ift} + \varepsilon_{ift}. \end{aligned} \quad (12)$$

The paper also presents split-sample versions by $StrongNCA_s$.

3.11 Model 8: Joint mobility of inventor pairs (Table 10)

For each bankruptcy event, form inventor pairs (i, j) employed by the same firm pre-bankruptcy. Define $MoveTogether_{ij} = 1$ if both inventors move to the same post-bankruptcy employer. The pair-level regression is:

$$\begin{aligned} MoveTogether_{ij} = & \alpha + \rho PairCodep_{ij} + \xi PairQuality_{ij} \\ & + \eta StrongNCA + \zeta Acquisition^{Strategic} + \kappa Liquidation + \varepsilon_{ij}, \end{aligned} \quad (13)$$

where $PairQuality_{ij}$ is alternatively omitted or measured via pair average citations, depending on the column.

4 Interpreting all figures and tables (what each one shows and why it matters)

4.1 Figure 1: Bankruptcies over time

What it is: frequency of bankruptcies by year (1980–2015) in the sample.

Why it matters: bankruptcies are staggered and occur in many years; this supports the DID logic in which not-yet-treated inventors serve as controls for currently treated inventors.

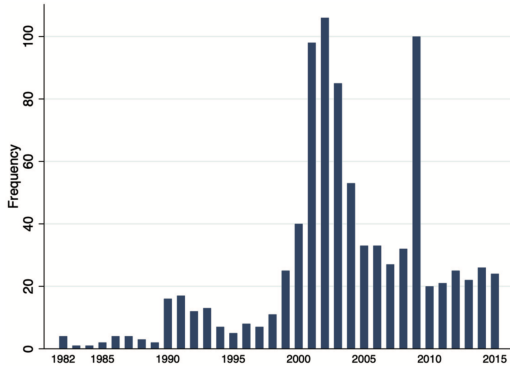


Figure 1
Corporate bankruptcies over time
This figure reports the frequency of corporate bankruptcies by year from 1980 to 2015, as they occur in our sample. Information on bankruptcy filings is from the UCLA-LoPucki Bankruptcy Research Database and the New Generation Research bankruptcy database.

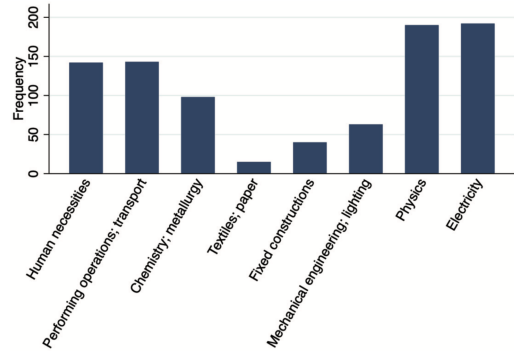


Figure 2
Corporate bankruptcies across industries
This figure shows the distribution of corporate bankruptcies by industry between 1980 and 2015, as they occur in our sample. The industry classification reflects the Cooperative Patent Classification (CPC) technology class and is based on the modal patent class for each bankrupt firm. Information on bankruptcy filings is from the UCLA-LoPucki Bankruptcy Research Database and the New Generation Research bankruptcy database.

4.2 Figure 2: Bankruptcies across industries

What it is: distribution of bankruptcies across industries.

Why it matters: bankruptcies are not confined to one narrow sector; this reduces concerns that results are driven by a single industry cycle and motivates technology-class-by-year controls.

4.3 Figure 3: Team production in innovation over time

What it is: average number of coauthors per patent over time.

Why it matters: teamwork is increasingly important in innovation. This motivates focusing on team-specific capital as a first-order object in modern production.

4.4 Figure 4: Firm characteristics prior to bankruptcy

Model: an event-time regression of firm characteristics Y_{ft} on treatment status interacted with event-time dummies:

$$Y_{ft} = \alpha + Treated_f \cdot \mathbf{T}'_{ft}\beta + \Gamma'\mathbf{Z}_{ft} + \varepsilon_{ft},$$

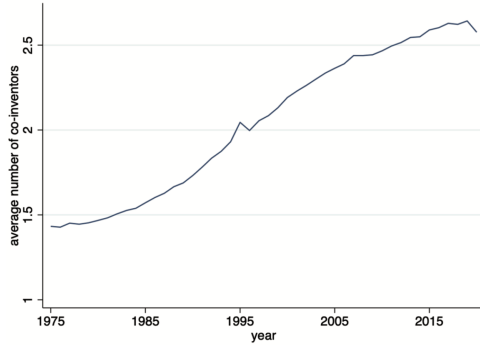


Figure 3
Team production in corporate innovation over time
This figure shows the evolution of the average number of coauthors per patent between 1976 and 2020. The data are from the PatentsView database.

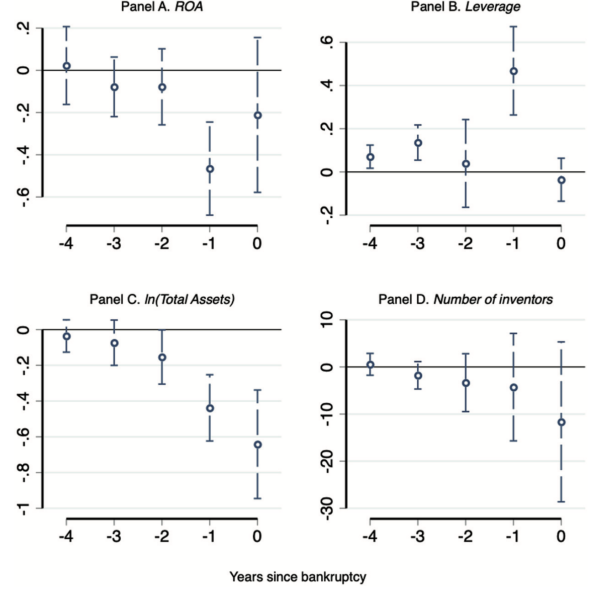


Figure 4
Firm characteristics as firms approach bankruptcy
This figure shows how firm characteristics evolve prior to the bankruptcy filing. We first estimate the following ordinary least squares regression model:

with firm FE and industry-by-year FE.

Interpretation: treated firms' ROA deteriorates and leverage rises as bankruptcy approaches; firm size and inventor counts exhibit systematic dynamics. This figure is mainly diagnostic: it documents pre-bankruptcy distress and helps interpret why team reallocation frictions may be severe near bankruptcy.

4.5 Figure 5: Team dependence and productivity around bankruptcy

Model: dynamic version (7) using only treated inventors and event-time dummies interacted with $TeamDep_i$.

Interpretation: coefficients are near zero pre-bankruptcy (supporting parallel trends in the interaction dimension), then become negative and persistent post-bankruptcy, especially for quality measures (citations-based). This is direct visual evidence consistent with a loss of team-specific capital after bankruptcy.

4.6 Table 1: Summary statistics

Panel A: main panel regressions (Tables 3–6, 8–9).

Key values to keep in mind:

- Mean $Post$ is about 0.36 (a large post-treatment region).
- Mean $TeamDep$ is about 0.49 (substantial reliance on teamwork).
- Mean $RemainTogether$ is about 0.55 (teams are often stable absent disruption, leaving room for large declines).

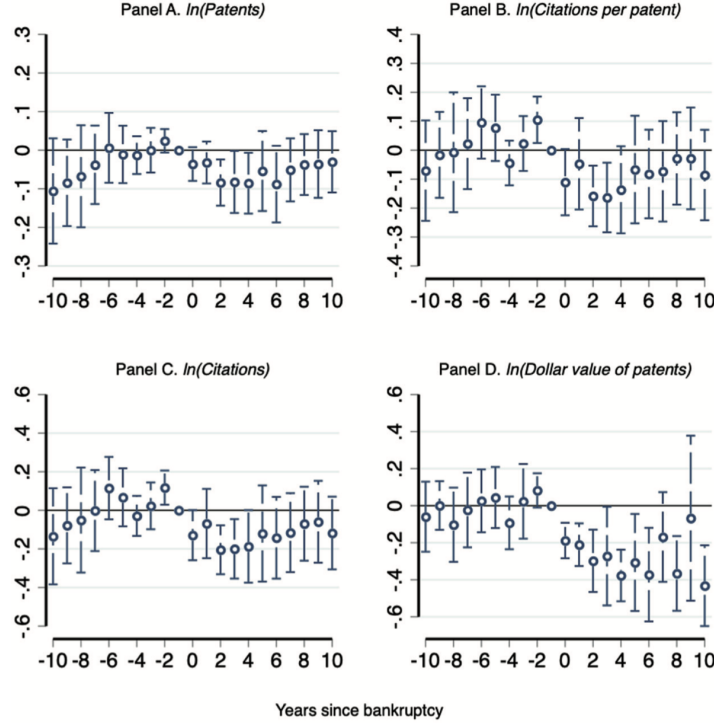


Figure 5
Team dependence and inventor productivity around corporate bankruptcies

Panel B: survival outcomes used in Table 7 (share still inventing 1–5 years post-bankruptcy declines monotonically).

4.7 Table 2: Treated vs never-treated inventors (selection diagnostics)

Panel A: technology classes. Treated and never-treated inventors are distributed across similar broad technology classes, with some differences in composition.

Panel B: inventor characteristics. Treated inventors are *not* low-quality: they patent slightly more per year, receive more citations, and have longer inventive careers (active years, lifetime patents/citations).

Interpretation: these comparisons reduce (but do not eliminate) concerns that bankrupt firms simply employ “weak” inventors; if anything, treated inventors look stronger.

4.8 Table 3: Bankruptcy reduces inventor productivity (baseline)

Panel A (Equation (3) and (4)):

- The average post-bankruptcy coefficient for $\ln(\text{Patents} + 1)$ is about -0.039 (roughly a 4% drop).
- The dynamic event-study shows declines starting in the filing year (t_0) and deepening through about three years post-bankruptcy.

Table 1
Summary statistics

A. Summary statistics for variables used in Tables 3, 4, 5, 6, 8, and 9			
	Obs.	Mean	SD
<i>ln(Patents)</i>	162,338	0.390	0.549
<i>ln(Citations per patent)</i>	162,338	0.799	1.284
<i>ln(Citations)</i>	162,338	0.943	1.502
<i>ln(Dollar value of patents)</i>	81,276	0.772	1.284
<i>Years of experience</i>	162,338	11.679	8.084
<i>Post-bankruptcy</i>	162,338	0.361	0.480
<i>Team dependence</i>	162,338	0.486	0.425
<i>Tenure</i>	162,338	6.757	5.108
<i>Change tech. class</i>	162,338	0.290	0.454
<i>t0</i>	162,338	0.065	0.247
<i>t1</i>	162,338	0.060	0.237
<i>t2</i>	162,338	0.054	0.226
<i>t3</i>	162,338	0.049	0.216
<i>t4</i>	162,338	0.043	0.203
<i>t5plus</i>	162,338	0.155	0.362
<i>t3plus</i>	162,338	0.247	0.431
<i>Noncompete enforcement index</i>	158,063	3.932	2.020
<i>Strong NCA enforcement</i>	158,063	0.595	0.491
<i>Independent exit</i>	140,424	0.602	0.489
<i>Acquisition by financial buyer</i>	140,424	0.048	0.214
<i>Acquisition by strategic buyer</i>	140,424	0.112	0.315
<i>Liquidation</i>	140,424	0.275	0.446
<i>Remain together</i>	127,983	0.546	0.498
<i>Past generality</i>	111,700	0.578	0.257

B. Summary statistics for variables used in Table 7			
	Obs.	Mean	SD
<i>Still inventing (1)</i>	12,089	0.729	0.444
<i>Still inventing (2)</i>	12,089	0.668	0.471
<i>Still inventing (3)</i>	12,089	0.608	0.488
<i>Still inventing (4)</i>	12,089	0.551	0.497
<i>Still inventing (5)</i>	12,089	0.498	0.5
<i>Team dependence</i>	12,089	0.497	0.441

C. Summary statistics for variables used in Table 10			
	Obs.	Mean	SD
<i>Move together</i>	2,155,490	0.010	0.098
<i>Pair codependence</i>	2,155,490	0.001	0.022
<i>Pair average citations</i>	2,155,490	0.141	4.017
<i>Strong NCA enforcement</i>	2,154,562	0.923	0.266
<i>Acquisition by strategic buyer</i>	2,145,461	0.011	0.104
<i>Liquidation</i>	2,145,461	0.630	0.483

This table shows summary statistics for the variables used in the regressions reported in Tables 3 to 10. The variable definitions are reported in the [appendix](#).

Table 2
Comparing inventors in bankruptcy firms to other inventors

A. Technology classes					
	Never-treated		Treated		
	Observations	Percentage	Observations	Percentage	
Necessities	1,306,626	16.4	10,484	7.9	
Transport	1,200,585	15.1	28,556	21.5	
Chem./metal.	999,554	12.6	16,908	12.7	
Textiles	67,323	0.8	2,041	1.5	
Construction	277,627	3.5	3,705	2.8	
Mech. engineering	574,309	7.2	17,521	13.2	
Physics	2,032,280	25.5	28,224	21.2	
Electricity	1,497,094	18.8	25,648	19.3	

B. Inventor characteristics					
	Never-treated			Treated	
	Observations	Mean	SD	Observations	Mean
<i>Patents</i>	7,957,055	0.673	1.641	133,087	0.753
<i>Citations</i>	7,957,055	14.661	97.572	133,087	18.621
<i>Citations per patent</i>	7,957,055	8.251	36.154	133,087	9.724
<i>Lifetime patents</i>	658,187	7.332	14.228	12,147	11.694
<i>Lifetime citations</i>	658,187	156.634	819.620	12,147	229.055
<i>Inventor ranking (patents)</i>	658,187	68.971	21.965	12,147	79.505
<i>Inventor ranking (citations)</i>	658,187	63.399	27.149	12,147	73.952
<i>Active years</i>	658,187	10.096	7.597	12,147	14.931
<i>Inventors per patent</i>	658,187	3.323	1.982	12,147	3.230

- For citations-per-patent, the average post coefficient is negative but less precisely estimated in the simplest specification; the dynamic effects are clearer.

Panel B (portability):

- Inventors who *change technology class* after bankruptcy experience a significantly smaller productivity decline (positive interaction).
- Inventors with more *general* pre-bankruptcy skills show weaker (less precise) mitigation.

Economic reading: portability helps; bankruptcy harms innovation, consistent with disruption and reallocation costs.

4.9 Table 4: Bankruptcy destabilizes teams

Outcome: *RemainTogether_{ift}* (team stability indicator).

Baseline (column 1): large negative effects in the filing year and year after ($t0, t1$), a weaker effect in year two, and little effect after year three (coefficients near zero), indicating a sharp team-disruption episode.

By bankruptcy outcome (columns 2–5):

- **Liquidation** magnifies team disruption (more negative interactions).
- **Strategic acquisition** attenuates disruption (positive interactions, especially around $t1$).

Table 3

Post-bankruptcy productivity of inventors

A				
	(1)	(2)	(3)	(4)
	<i>ln(Patents)</i>	<i>ln(Citations per patent)</i>	<i>ln(Patents)</i>	<i>ln(Citations per patent)</i>
<i>Post-bankruptcy</i>	−0.039** (0.017)	−0.031 (0.024)		
<i>t0</i>			−0.030*** (0.010)	−0.057** (0.023)
<i>t1</i>			−0.043*** (0.013)	−0.046* (0.024)
<i>t2</i>			−0.048*** (0.015)	−0.059** (0.026)
<i>t3</i>			−0.064*** (0.026)	−0.063** (0.033)
<i>t4</i>			−0.061 (0.037)	−0.056 (0.054)
<i>t5plus</i>			−0.074 (0.047)	−0.088 (0.065)
Technology class-by-year FE	Yes	Yes	Yes	Yes
Years of experience FE	Yes	Yes	Yes	Yes
Inventor FE	Yes	Yes	Yes	Yes
Observations	162,338	162,338	162,338	162,338
Adjusted R^2	.322	.283	.322	.283
B				
	(1)	(2)	(3)	(4)
	<i>ln(Patents)</i>	<i>ln(Citations per patent)</i>	<i>ln(Patents)</i>	<i>ln(Citations per patent)</i>
<i>Post-bankruptcy</i>	−0.064*** (0.021)	−0.059** (0.027)	−0.058** (0.026)	−0.094* (0.050)
<i>Post-bankruptcy * Change tech. class</i>	0.079*** (0.013)	0.085*** (0.022)		
<i>Post-bankruptcy * Past generality</i>			0.017 (0.031)	0.056 (0.057)
Technology class-by-year FE	Yes	Yes	Yes	Yes
Years of experience FE	Yes	Yes	Yes	Yes
Inventor FE	Yes	Yes	Yes	Yes
Observations	162,338	162,338	111,698	111,698
Adjusted R^2	.323	.283	.322	.283

Table 4

Team stability after bankruptcies

	(1)	(2)	(3)	(4)	(5)	(6)
Remain together						
<i>t0</i>	−0.202** (0.084)	−0.136* (0.070)	−0.331** (0.151)	−0.218** (0.097)	−0.235** (0.100)	−0.123** (0.053)
<i>t1</i>	−0.247*** (0.090)	−0.198*** (0.065)	−0.386** (0.160)	−0.283*** (0.100)	−0.306*** (0.102)	−0.146** (0.063)
<i>t2</i>	−0.149 (0.103)	−0.107 (0.081)	−0.249 (0.195)	−0.199* (0.116)	−0.211* (0.120)	−0.041 (0.074)
<i>t3plus</i>	0.040 (0.097)	0.096 (0.082)	−0.065 (0.195)	0.009 (0.112)	−0.011 (0.115)	0.119* (0.071)
<i>t0 * Liquidation</i>		−0.317* (0.164)				
<i>t1 * Liquidation</i>		−0.314* (0.166)				
<i>t2 * Liquidation</i>		−0.298 (0.199)				
<i>t3plus * Liquidation</i>		−0.340* (0.187)				
<i>t0 * Independent exit</i>			0.182 (0.159)			
<i>t1 * Independent exit</i>			0.172 (0.158)			
<i>t2 * Independent exit</i>			0.104 (0.189)			
<i>t3plus * Independent exit</i>			0.131 (0.181)			
<i>t0 * Acquisition by financial buyer</i>				−0.010 (0.118)		
<i>t1 * Acquisition by financial buyer</i>				0.038 (0.128)		
<i>t2 * Acquisition by financial buyer</i>				0.224 (0.177)		
<i>t3plus * Acquisition by financial buyer</i>				0.139 (0.145)		
<i>t0 * Acquisition by strategic buyer</i>					0.145 (0.129)	
<i>t1 * Acquisition by strategic buyer</i>					0.234* (0.124)	
<i>t2 * Acquisition by strategic buyer</i>					0.218 (0.133)	
<i>t3plus * Acquisition by strategic buyer</i>					0.225** (0.110)	
<i>t0 * Strong NCA enforcement</i>						−0.137* (0.082)
<i>t1 * Strong NCA enforcement</i>						−0.169** (0.077)
<i>t2 * Strong NCA enforcement</i>						−0.190** (0.082)
<i>t3plus * Strong NCA enforcement</i>						−0.129* (0.066)
Inventor FE	Yes	Yes	Yes	Yes	Yes	Yes
Technology class-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Years of experience FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	127,465	108,956	108,956	108,956	108,956	121,730
Adjusted R^2	.349	.372	.363	.360	.362	.348

- Independent exit and financial buyer outcomes show weaker/less consistent mitigation.

Mobility restrictions (column 6): stronger noncompete enforceability reduces team stability around bankruptcy, consistent with constrained re-matching/reallocation.

4.10 Table 5: Team-dependent inventors are hit harder (core test)

Panel A (Equation (6)):

- Before bankruptcy, *TeamDep* is positively associated with productivity (team inventors are more productive).
- Post-bankruptcy, the interaction $Post \times TeamDep$ is strongly negative: for an inventor with $TeamDep = 1$, the post-bankruptcy drop in $\ln(\text{Patents} + 1)$ is about $-0.016 - 0.061 \approx -0.077$ (about 7–8%), and the drop in $\ln(\text{Citations per patent} + 1)$ is about $-0.001 - 0.099 \approx -0.100$ (about 10%).
- The quality margin (citations per patent) is especially sensitive to team disruption.

Panel B (dynamic interactions): The negative interaction is concentrated after bankruptcy and persists for several years, which supports a causal interpretation via team-capital loss rather than instantaneous mechanical effects.

Table 5
Team dependence and the post-bankruptcy productivity of inventors

A				
	(1)	(2)	(3)	(4)
	<i>ln(Patents)</i>	<i>ln(Citations per patent)</i>	<i>ln(Patents)</i>	<i>ln(Citations per patent)</i>
Post-bankruptcy	-0.045*** (0.016)	-0.047** (0.023)	-0.016 (0.014)	-0.001 (0.024)
Team dependence	0.029 (0.022)	0.049 (0.035)	0.053** (0.024)	0.088** (0.043)
Post-bankruptcy * Team dependence			-0.061*** (0.013)	-0.099*** (0.031)
Bankruptcy firm FE	Yes	Yes	Yes	Yes
Technology class-by-year FE	Yes	Yes	Yes	Yes
Years of experience FE	Yes	Yes	Yes	Yes
Observations	164,456	164,456	164,456	164,456
Adjusted R ²	.111	.178	.112	.179

B				
	(1)	(2)	(3)	(4)
	<i>ln(Patents)</i>	<i>ln(Citations per patent)</i>	<i>ln(Patents)</i>	<i>ln(Citations per patent)</i>
Post-bankruptcy * Team dependence	-0.047*** (0.015)	-0.104** (0.041)		
t0 * Team dependence			-0.024* (0.014)	-0.138*** (0.038)
t1 * Team dependence			-0.020 (0.014)	-0.075 (0.056)
t2 * Team dependence			-0.072*** (0.013)	-0.187*** (0.044)
t3 * Team dependence			-0.070*** (0.020)	-0.192*** (0.047)
t4 * Team dependence			-0.074*** (0.021)	-0.165*** (0.055)
t5plus * Team dependence			-0.044* (0.024)	-0.092* (0.054)
Bankruptcy firm-by-year FE	Yes	Yes	Yes	Yes
Technology class-by-year FE	Yes	Yes	Yes	Yes
Inventor FE	Yes	Yes	Yes	Yes
Years of experience FE	Yes	Yes	Yes	Yes
Observations	157,950	157,950	157,950	157,950
Adjusted R ²	.355	.344	.355	.345

Table 6
Team dependence and the post-bankruptcy productivity of inventors: Robustness tests

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	<i>ln(Patents)</i>	<i>ln(Patents)</i>	<i>ln(Citations)</i>	<i>ln(Dollar value of patents)</i>	<i>ln(Patents)</i>	<i>ln(Patents)</i>	<i>ln(Patents)</i>	<i>Patents</i>
Post-bankruptcy * Team dependence	-0.076** (0.031)	-0.054*** (0.016)	-0.113*** (0.038)	-0.240*** (0.056)	-0.043*** (0.015)	-0.043*** (0.015)	-0.056*** (0.016)	-0.285*** (0.070)
Post-bankruptcy * Tenure					(0.001)			
Bankruptcy firm-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Inventor FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Technology class-by-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Years of experience FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	64,708	54,054	157,950	79,090	157,950	8,107,175	111,653	146,004
Adjusted R ²	.299	.367	.357	.370	.355	.337	.365	

4.11 Table 6: Robustness

Key robustness lessons:

- Effects persist for **star inventors** (top decile of patenters) and for firms with **low R&D intensity**.
- Using alternative outcomes (total citations; patent dollar value) yields consistent negative post × team dependence effects.
- Adding **tenure** and its interaction with *Post* does not remove the team-dependence effect.
- Including inventors who never experience bankruptcy in the control group yields similar patterns.
- Using a **Poisson count model** for patents delivers consistent conclusions, addressing functional-form concerns.

4.12 Table 7: Career continuation (selection)

Team-dependent inventors are less likely to keep inventing after bankruptcy. The coefficient is most negative for the short horizon (still inventing within 1 year), and attenuates as the horizon expands to 5 years.

Interpretation: bankruptcy can induce exit from innovative activity, and the exit margin is more pronounced for team-dependent inventors. This implies that productivity effects computed conditional on continued patenting may understate total welfare losses.

4.15 Table 10: Direct evidence of joint team mobility (pair-level)

Pairs with stronger prior collaboration and higher-quality joint output are more likely to move to the *same* post-bankruptcy employer:

- **Pair codependence** has a large positive coefficient: more tightly linked pairs are more likely to relocate together.
- **Pair average citations** also predicts joint relocation: *productive* pairs are preserved.
- **Strong noncompete enforcement** reduces joint mobility.
- **Strategic acquisitions** increase, while **liquidations** decrease, the likelihood of joint mobility.

Interpretation: the labor market and market for corporate control help preserve team-specific human capital, but contractual mobility frictions impede it.

5 Summary

1. Bankruptcy reduces inventors' output and patent impact (Table 3), and the decline begins around filing and persists.
2. Teams become less stable around bankruptcy (Table 4): disruption is concentrated near t_0 and t_1 .
3. The productivity decline is *much larger* for inventors whose pre-bankruptcy production relied on teamwork (Table 5; Figure 5).
4. Results are robust across samples, outcomes, and functional forms (Table 6).
5. Bankruptcy also induces inventor exit from patenting, especially among team-dependent inventors (Table 7).
6. Resolution matters: strategic acquisitions preserve teams and mitigate losses, while liquidation worsens (Tables 4 and 8).
7. Mobility restrictions (noncompete enforceability) exacerbate both team disruption and productivity losses (Tables 4 and 9).
8. Pair-level evidence shows that cohesive and productive inventor pairs are more likely to move together post-bankruptcy (Table 10), suggesting active preservation of team-specific capital when frictions allow.

6 Conclusion

6.1 What the paper concludes

This paper concludes that **corporate bankruptcy reduces innovation productivity partly by destroying team-specific human capital**. Bankruptcy is not only a financial event; it is also a

reallocation shock that can break up inventor teams and disrupt collaborative routines. Because modern innovation is increasingly produced by teams, these disruptions can generate persistent losses in both the **quantity** and **quality** of inventive output.

Empirically, the paper combines corporate bankruptcy events with inventor–firm panel data and patents, and uses inventor fixed effects plus rich time-varying controls to identify within-inventor productivity changes around bankruptcy. The evidence supports a coherent mechanism:

- (i) bankruptcies reduce inventor output and impact,
- (ii) bankruptcies substantially destabilize inventor teams,
- (iii) inventors who are more **team-dependent** experience larger productivity declines,
- (iv) outcomes that preserve teams (especially **strategic acquisitions**) mitigate the losses,
- (v) mobility frictions (strong **noncompete enforceability**) exacerbate both team disruption and productivity losses,
- (vi) and the labor market partially recognizes team capital: highly collaborative/high-quality inventor pairs are more likely to move together.

6.2 Key empirical conclusions to remember (with the paper’s headline magnitudes)

The most exam-relevant results can be summarized as:

- (1) **Average productivity falls after bankruptcy (Table 3).** The post-bankruptcy indicator is negative for $\ln(\text{Patents} + 1)$ (about -0.039), implying roughly a 4% decline in patent output. Event-time coefficients show declines beginning around the filing year (t_0) and persisting for multiple years.
- (2) **Teams break apart sharply around filing (Table 4).** The probability that an inventor remains with (most of) their pre-existing team drops substantially in the filing year and the following year (e.g., about -0.20 at t_0 and -0.25 at t_1 in the baseline specification). This indicates that the core disruption is concentrated around the bankruptcy episode, consistent with a discrete shock to team continuity.
- (3) **Team dependence predicts larger post-bankruptcy losses (Table 5, Figure 5).** The interaction $Post \times TeamDependence$ is strongly negative for both patent quantity and quality (about -0.061 for $\ln(\text{Patents} + 1)$ and about -0.099 for $\ln(\text{Citations per patent} + 1)$). This means that inventors whose pre-bankruptcy production relied more on within-firm collaboration suffer larger productivity declines.
- (4) **Resolution matters: liquidation worsens, strategic acquisition mitigates (Table 4 and Table 8).** Liquidation is associated with additional team disruption and larger team-dependent productivity losses. In contrast, acquisition by a **strategic buyer** (an operating

company) significantly mitigates team disruption and offsets the team-dependent productivity decline (positive interaction terms in Table 8). Intuitively, strategic buyers are more likely to preserve (or reconstruct) teams and their complementary assets.

- (5) **Mobility restrictions exacerbate the losses (Table 9).** In states with stronger non-compete enforceability, the negative $Post \times TeamDependence$ effects are much larger (e.g., about -0.074 for $\ln(Patents + 1)$ and about -0.168 for $\ln(Citations \text{ per patent} + 1)$ in the high-enforcement subsample). In the pooled specification, the triple interaction $Post \times TeamDependence \times StrongNCA$ is negative, consistent with mobility frictions preventing teams from re-forming or moving together.
- (6) **The labor market partially preserves team-specific capital via joint mobility (Table 10).** Inventor pairs with stronger prior collaboration (*pair codependence*) and higher-quality joint output (*pair average citations*) are significantly more likely to move to the same post-bankruptcy employer. Strong noncompete enforceability reduces joint mobility, strategic acquisitions increase it, and liquidation reduces it.

6.3 Why bankruptcy destroys innovation through teams

The paper’s mechanism can be summarized using a simple conceptual production view.

- 1) Team-specific human capital as an input.** Think of inventor productivity as depending on both individual skill and team-specific complementarities:

$$y_{ift} = \underbrace{a_i}_{\text{individual skill}} + \underbrace{\theta H_{ift}}_{\text{team-specific human capital}} + \underbrace{u_{ift}}_{\text{other shocks}}, \quad (14)$$

where H_{ift} captures routines, match quality, and complementarities that are created by repeated collaboration.

- 2) Bankruptcy as a discrete shock to the team-capital stock.** Bankruptcy increases the probability of separation, re-matching, and organizational reconfiguration. If team-specific capital is partly *relationship-specific* and costly to rebuild, bankruptcy can reduce H_{ift} sharply:

$$H_{ift}^{post} = (1 - \Delta_f) H_{ift}^{pre}, \quad \Delta_f > 0, \quad (15)$$

which generates a decline in y_{ift} even if the inventor’s underlying ability a_i is unchanged.

- 3) Why team dependence matters.** The measure *Team dependence* proxies for how strongly an inventor’s output loads on H_{ift} . Inventors with higher team dependence have a larger θ in (14), so when bankruptcy reduces H_{ift} , they suffer larger productivity losses. This is exactly what the negative $Post \times TeamDependence$ estimates capture.

4) Why strategic acquisitions mitigate, and liquidations worsen. Strategic buyers are more likely to transfer complementary assets, maintain project continuity, and re-employ teams together, so they reduce Δ_f in (15). Liquidation does the opposite: it fragments assets and relationships, making it difficult for teams to remain intact or re-form.

5) Why mobility frictions exacerbate the loss. When noncompete enforcement is strong, coordinated team moves are harder. This increases re-matching frictions and raises Δ_f , making it less likely that team-specific capital is preserved through joint mobility or rapid re-formation in a new firm.

6.4 Takeaway

Corporate bankruptcies reduce innovation not only by financial distress but also by breaking inventor teams: team stability drops sharply at filing, team-dependent inventors suffer substantially larger declines in output and patent impact, strategic acquisitions preserve teams and mitigate losses, liquidation worsens them, and mobility restrictions such as strong noncompete enforcement amplify the destruction of team-specific human capital.